Problems: Kernel Density Estimation

You may use code to solve these problems. You may then sketch what your plots show, or add the plots to this homework and submit a pdf digitally (email).

Problem 1. Suppose the data are `[1, 1.75, 2, 3, 3.25, 4]`.

- a. Sketch the empirical cumulative distribution function.
- b. Sketch the uniform kernel density estimator for a bandwidth of `h = 0.2`
- c. Compute the Silverman bandwidth for the uniform kernel using `h = 1.84 * SD * n**(-0.2)`.
- d. Sketch the uniform kernel density estimator for the Silverman bandwidth.
- e. How are b and d similar or different?

Problem 2. Suppose the data `X = [-1, -0.5, 0, 0.5, 1]` were drawn from a standard Normal distribution.

- a. Using the uniform kernel with `h = 0.6`, compute the realized KDE at `x = 0`, `f_hat_h(0)`, directly from these five numbers.
- b. Now compute `E[f_hat_h(0)]` at the same `h = 0.6`.
- c. Calculate the true density at `x=0`, using the Normal distribution.
- d. Are b and c equal? What does this mean about bias? (Keep in mind, truly proving anything about bias would require us to compare `E[f_hat_h]` to the Normal distribution for all *x*; calculating this at one point is only an indication of what is going on that you can extrapolate from.)
- e. Recompute `E[f_hat_h(0)]` from (b) at a much smaller bandwidth, `h = 0.1`. What happens to the gap between `E[f_hat_h(0)]` and `f(0)` as `h` shrinks?
- f. Is `f_hat_h` a consistent estimator? Explain your reasoning.
- g. What are the main considerations in choosing a bandwidth value `h`?

Problem 3. Suppose the data are `X = [5, 7, 7, 8, 10]`, and fix the bandwidth at `h = 1.5` for both kernels below.

- a. Compute the uniform-kernel KDE, `f_hat_h(x)`, at `x = 6.0`, `x = 6.5`, `x = 7.0`, `x = 8.0`, and `x = 8.5`.
- b. Compute the Gaussian-kernel KDE at the same five values of `x`, using the same `h = 1.5`.
- c. Your part (a) values should jump abruptly between some of these five points, while your part (b) values should change smoothly. Identify where the biggest jump in (a) happens, and explain in terms of the two kernel *shapes* -- not just "one is smoother" -- why that jump exists in the uniform case but not the Gaussian case.
- d. Why might you prefer one kernel over the other in practice?

Problem 4. Load `./data/heart_failure_clinical_records_dataset.csv`.

- a. Plot a
- kernel density estimate of `ejection_fraction`.
- b. Use Pandas' `df.sample(frac=1.0, replace=True)` to resample the data 15 times with replacement, plotting the kernel density estimtes for each sample.
- c. For what values of `ejection_fraction` is the original plot reliable? For which values is there noticeable variation in your plots of the resampled data?